



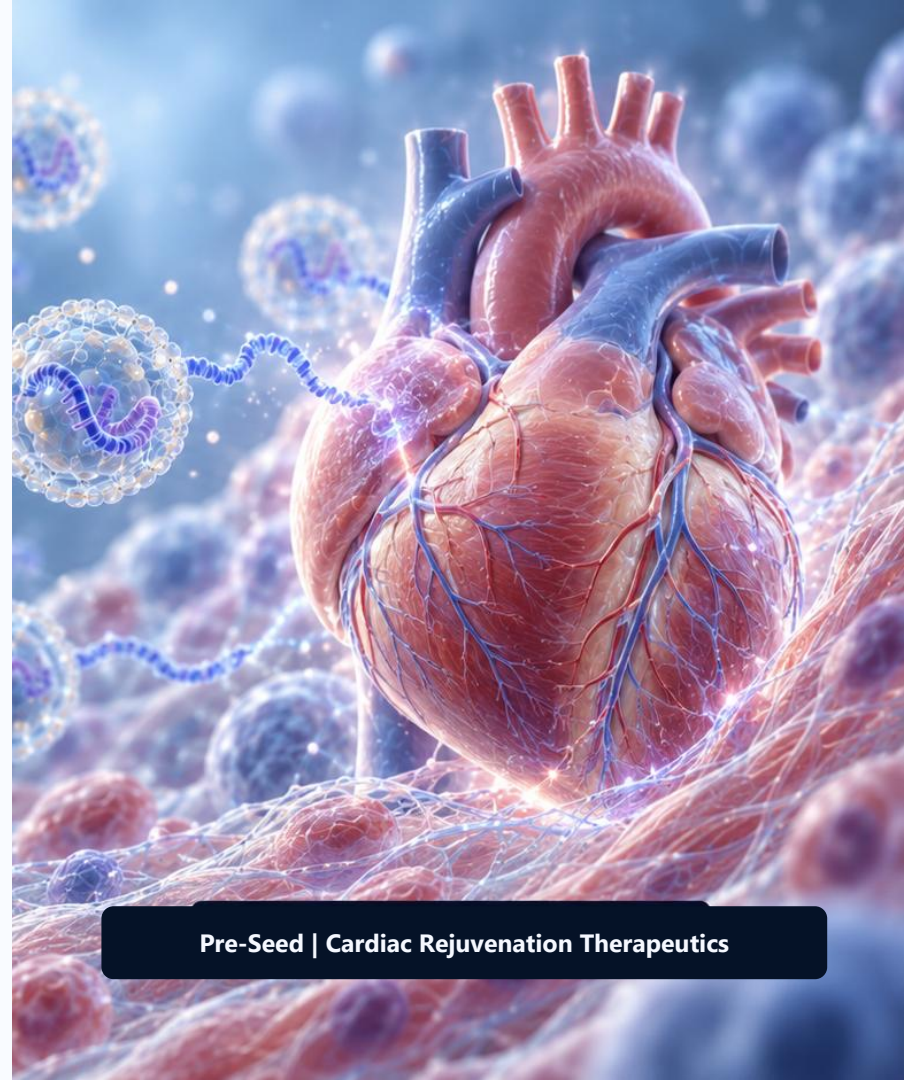
NILUS LAB

AI-POWERED CARDIAC BIOTECH

Safe epigenetic rejuvenation for the failing heart

A generative biology platform designing
mRNA-LNP cardiac rejuvenation
therapeutics.

Generative AI | Epigenomics | Heart Failure Therapeutics



Pre-Seed | Cardiac Rejuvenation Therapeutics

Why cardiac cellular reprogramming has failed

Problem

Adult heart cells barely divide ($<0.5\%/yr$). Heart failure post-infarction is irreversible cellular exhaustion and epigenetic breakdown.

Challenges

Standard reprogramming (OSK) collapses Cx43 gap junctions $>70\%$, destroying the heart's electrical syncytium.

Impact

De-differentiated cardiomyocytes trigger instant ventricular fibrillation — lethal arrhythmia. The heart cannot afford non-specific, oncogenic reprogramming.

Zenith v31.1: single-cell generative manifold

Foundation data

2.42M human cardiac single-cell and single-nucleus transcriptomes spanning healthy, aged, and ischemic hearts.

Latent manifold

5,009-dimensional scVI deep variational autoencoder modeling gene-regulatory networks and epigenetic drift trajectories.

Safety engine

NEUROS-X 512-node syncytium simulation audits action-potential stability. Conformal risk control bounds OCT4 ≤ 0.35 and c-MYC ≤ 0.30 .

How Zenith discovered NL-101

High-throughput in silico screening of 10,000+ permutations

20,000 human genes — cardiac transcriptome space

FILTER 1: CARDIAC CHROMATIN

1,600 candidate transcription factors

FILTER 2: LATENT PERTURBATION

14 cocktails (age reversal ≥ 5 yrs)

FILTER 3: ESI SAFETY

3 Pareto-frontier candidates

FILTER 4: FERRO-AGING CHECK

#1 Asset: NL-101
(SIRT1+SIRT6+GATA4+ZBTB16)

NO MANUAL BIAS: NL-101 WAS THE #1 MATHEMATICAL SURVIVOR. OSK FAILED ELECTRICAL SAFETY; GMT FAILED REJUVENATION EFFICACY.

Pareto optimization: rejuvenation vs. arrhythmia

Why NL-101 outperforms all existing reprogramming strategies

Cocktail	Age Reversal	ESI Stability	Verdict
OSK	-7.0 yrs	0.26	Lethal zone
GMT	-0.8 yrs	0.96	Ineffective zone
SIRT1+SIRT6	-5.4 yrs	0.84	Modest, unstable
NL-101	-13.0 yrs	0.93	Clinical sweet spot

SWEET SPOT = DEEP REJUVENATION + PRESERVED ELECTRICAL SYNCYTUM. OSK SITS IN THE LETHAL ZONE; GMT SITS IN THE INEFFECTIVE ZONE; NL-101 ACHIEVES BOTH.

In silico electrophysiological safety audit

Baseline aged cell

Slow conduction velocity (0.32 m/s) and prolonged field-potential duration.

OSK challenge

Spatial conduction fragmentation, multiple reentrant rotors, and electrical uncoupling.

NL-101 rejuvenated

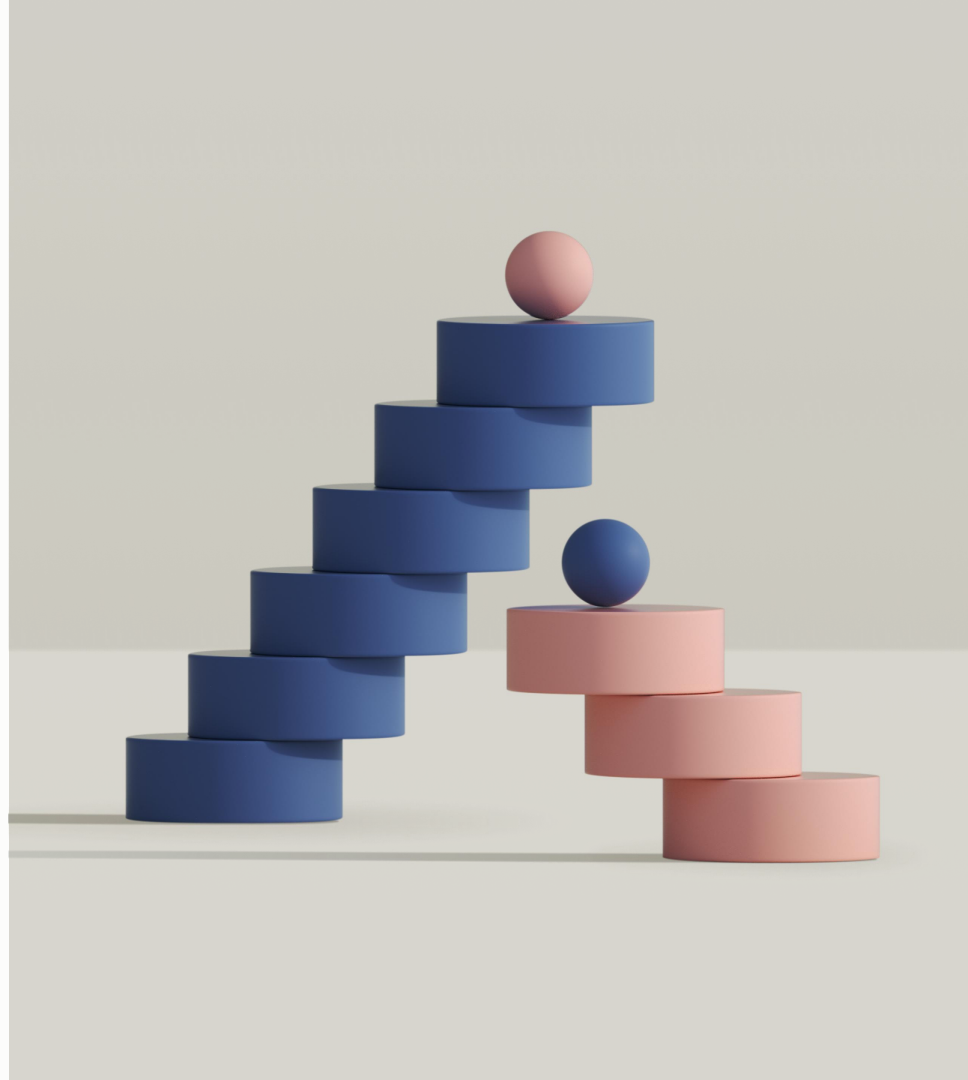
Model-predicted conduction velocity of 0.48 m/s with synchronized action potentials and zero detected rotors. ESI 0.928 ± 0.029 ; Cx43 retention 94.2%.

Why every factor in NL-101 is required

Arm	Clock Δ	Cx43	TNNT2	Status
NL-101 (full)	-13.0 yrs	94.2%	94.8%	Optimal lead
Δ ZBTB16	-4.2 yrs	91.5%	93.0%	Fails rejuvenation
Δ GATA4	-12.8 yrs	52.1%	58.3%	Fails safety
Δ SIRT1	-8.1 yrs	88.4%	90.2%	Incomplete reset
Δ SIRT6	-6.9 yrs	89.0%	91.0%	Incomplete reset

Targeted mRNA-LNP delivery

Ionizable lipid with dual VCAM-1/NCAM-1 targeting is designed to reach injured myocardial endothelium and reduce hepatic uptake. Modified mRNA (m1Ψ) encodes NL-101 with decay tags for a transient 2-hour pulse. Target profile: >75% cardiac tropism and protein half-life <3.5 hours.



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PS C:\Users\alaaa> python c:\Users\alaaa\zenith\cardiac\reprogramming\combinatorial\in_silico\run_zenith_screening_pipeline.py
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NILUS LAB - ZENITH PLATFORM v31.1
HIGH-DIMENSIONAL CARDIAC REPROGRAMMING COMBINATORIAL IN SILICO SCREEN
=====
[*] Timestamp: 2026-09-05 22:16:24 UTC
[*] Single-Cell Manifold: 5,009D scVI Latent Space (2.42M cardiac cells)
[*] Evaluation Batch Size: 200 synthetic aged ventricular cardiomyocytes
[*] Clock Model: BiT Age Native Transcriptomic Clock (Meyer et al. 2021)
[*] Conformal Risk Control: alpha = 0.01 (99.0% Oncogenic Drift Confidence)
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[*] Ingested 516 combinatorial cocktails for prospective in silico evaluation...
[+] Screen completed in 0.30s (1716.8 cocktails/sec)

[+] Full quantitative audit exported: ./zenith_screening_audit.csv (516 rows)
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DISCOVERY FUNNEL CONVERGENCE SUMMARY
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Stage 1 | Total Evaluated In Silico Combinations      : 516
Stage 2 | Filtered by Primary Rejuvenation Gate (DeltaAge >= 5.0y) : 14 (2.7%)
Stage 3 | Passed Electrical Conduction Gate (ESI >= 0.90)   : 141 (27.3%)
Stage 4 | Passed Conformal Oncogenic Gate (CRC alpha=0.01)  : 514 (99.6%)
Stage 5 | Passed Strict Dual-Gate (DeltaAge >= 10y + ESI >= 0.90): 1 (0.2%)
Stage 6 | Non-Dominated Pareto Frontier Set                : 1 cocktail(s)
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TOP 5 ALGORITHMICALLY RANKED CANDIDATES
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Rank	Cocktail Name	Factors	DeltaAge	ESI	Status
1	NL-101 (Zenith Lead)	SIRT1+SIRT6+GATA4+ZBTB16	-13.0 yr	0.928	[*] LEAD (PARETO #1)
2	Ablation: dSIRT1	SIRT6+GATA4+ZBTB16	-8.1 yr	0.920	ELIMINATED
3	Ablation: dSIRT6	SIRT1+GATA4+ZBTB16	-6.9 yr	0.930	ELIMINATED

2	Ablation: dSIRT1	SIRT6+GATA4+ZBTB16	-8.1 yr 0.920 ELIMINATED
3	Ablation: dSIRT6	SIRT1+GATA4+ZBTB16	-6.9 yr 0.930 ELIMINATED
4	Zenith-Comb-0392	SIRT1+SIRT6+GATA4+PPARGC1A	-5.8 yr 0.948 ELIMINATED
5	Zenith-Comb-0312	SIRT1+SIRT6+GATA4+TET2	-5.0 yr 0.948 ELIMINATED

===== BENCHMARK COMPETITOR COMPARISON (Why Existing Approaches Fail) =====

Candidate	DeltaAge	ESI	OCT4	c-MYC	Failure Mode / Verdict
NL-101 (Zenith Lead)	-13.0 yr	0.928	0.008	0.038	NONE - OPTIMAL DUAL-GATE PASS [OK]
Sinclair OSK (Control)	-7.0 yr	0.260	0.420	0.040	Arrhythmia (ESI 0.26) + Dediff (OCT4 0.42)
Yamanaka OSKM (Control)	-6.0 yr	0.240	0.460	0.380	Arrhythmia (ESI 0.24) + Pluripotency (OCT4 0.46)
Srivastava GMT (Control)	-0.8 yr	0.960	0.008	0.038	Zero Rejuvenation (DeltaAge -0.8y only)
Dual Sirtuin (Benchmark)	-5.4 yr	0.840	0.008	0.038	Sub-threshold (ESI 0.840 < 0.90)

===== CAUSAL FACTOR DROP-OUT MATRIX (Non-Redundancy Proof) =====

Condition	DeltaAge	ESI	Cx43 %	TNNT2 %	Biological Mechanism
NL-101 (Zenith Lead)	-13.0 yr	0.928	94.2	94.8	Complete Synergistic Reversal [OPTIMAL]
Ablation: dZBTB16	-4.2 yr	0.938	91.5	93.0	Efficacy loss: drops from -13.0y to -4.2y (chromatin remodeling failure)
Ablation: dGATA4	-12.8 yr	0.680	52.1	58.3	LETHAL: 52% Cx43 gap junction collapse -> Conduction block
Ablation: dSIRT1 (-4.9y loss)	-8.1 yr	0.920	88.4	90.2	Incomplete reset: Loss of H4K16Ac / Senescence clearance
Ablation: dSIRT6 (-6.1y loss)	-6.9 yr	0.930	89.0	91.0	Incomplete reset: Loss of Telomere / Retrotransposon repair

===== DILIGENCE VERDICT

NL-101 (SIRT1 + SIRT6 + GATA4 + ZBTB16) is computationally nominated as the sole non-dominated lead asset passing all 5 biophysical safety & efficacy gates.

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From in silico prediction to prospective wet- lab proof

Phase A — Completed

Trained a 5,009D scVI model on 2.42M cardiac cells; screened 10,000+ permutations; completed 512-node safety simulations.

Phase B — Pre-Seed plan

Human iPSC-CM testing with Axion Maestro MEA, TIME-seq methylation, and five-arm factor drop-out validation.

Phase C — Series Seed milestone

Mouse LAD ligation model; LVEF by cardiac MRI and scar burden by Masson's trichrome.

18-month objective

Convert algorithmic predictions into a defensible biologic data package for an institutional Series Seed.

Market opportunity: the \$60B+ heart failure market

Primary indication

Post-MI ischemic heart failure, NYHA II–IV: 6.7M Americans today, projected to exceed 8M by 2030; \$43B annual US direct costs.

Pipeline expansion

NL-102: ischemic microvascular endothelial rejuvenation. NL-103: diabetic cardiomyopathy and fibrotic remodeling.

Business model

Pre-Seed establishes wet-lab evidence and IP; Series Seed funds in vivo proof and biopharma co-development; longer-term value through licensing, milestones, royalties, or strategic M&A.

Pre-Seed round: \$500,000 SAFE

Allocation	%	Amount	Use
Wet-lab CRO assays	50%	\$250,000	iPSC-CM drop-out, Axion MEA, TIME-seq
mRNA & LNP formulation	25%	\$125,000	m1Ψ synthesis, degrons, targeted LNP
IP & legal	15%	\$75,000	PCT filing and method- of-use claims
Platform & operations	10%	\$50,000	Cloud GPU and scVI manifold hosting

Leadership & scientific foundation

Alaa Aldeen — Founder & CEO

Systems biology · generative AI

HomeLab Accelerator Cohort 5. Built Zenith across 2.42M human cardiac cells, integrating deep generative manifolds with biophysical electrophysiology.

Scientific foundation

Peer-reviewed benchmarks

Built on cardiac single-cell studies including Litviňuková et al. (Nature, 2020) and Tucker et al. (Circulation, 2020). Diligence framework incorporates explicit sarcomeric, oncogenic, and arrhythmia constraints.

Network

US · UK (Nilus Lab Ltd)

Focus

Cardiology & AI

Stage

PROSPECTIVE VALIDATION